SimaX
Database Backup

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● Runtime running
Accepted jobs and schedules keep running when this window is closed.

Protection overview

Latest usable recovery point per source. Backup, integrity and restore-test evidence are tracked separately: a finished backup is not proof of recovery.

Restore | New backup policy

Needs attention

■ Required copy failed

ledger-prod

Run `run_7f3a` exported, but the required S3 copy was not finalized. Only 1 of 2 required copies exists for the 19 Oct 03:00 snapshot; the last complete recovery point is 18 Oct 03:00.

Inspect run and retry safely

▲ Observation gap

billing-prod

Table-change trigger lost change history access at 06:42. Changes since then are not being observed; schedule-based backups continue.

Review trigger prerequisites

▲ Schedule overdue

reports-dev

Expected daily run on 18 Oct and 19 Oct did not start. Last usable recovery point: 15 Oct 02:00.

See why it did not run

Sources

Times in Asia/Jakarta (UTC+07:00) · Mon 19 Oct 2026, 11:08

SOURCE	LATEST RECOVERY POINT	BACKUP	INTEGRITY	RESTORE TESTED	REQUIRED COPIES	STATUS
orders-prod PostgreSQL 17 PROD	19 Oct 03:00 8 h old	Finalized	Verified	16 Oct `drill_0412`	2 of 2	● Protected
billing-prod MySQL 8.4 PROD	19 Oct 02:00 9 h old	Finalized	Verified	9 Oct 10 days ago	2 of 2	▲ Degraded Change observation gap
ledger-prod Oracle Database 19c PROD	18 Oct 03:00 32 h old	Exported, copy failed	Not verified	Never	1 of 2	■ At risk Required copy failed
crm-staging SQL Server 2022 STAGING	19 Oct 04:30 6 h old	Finalized	Verified	Never	1 of 1	○ Not restore-tested
reports-dev PostgreSQL 16 DEV	15 Oct 02:00 4 days old	Finalized	Verified	Never	1 of 1	▲ Schedule overdue

Recent runs

RUN	SOURCE	STARTED BY	POLICY REVISION	STAGE REACHED	RESULT
`run_7f3a`	ledger-prod	Schedule · 03:00	`nightly-ledger r6`	Transfer to S3	■ Failed
`run_7f39`	crm-staging	Webhook · event evt_218	`crm-on-deploy r2`	Finalized and verified	● Succeeded
`run_7f37`	orders-prod	Schedule · 03:00	`nightly-orders r11`	Finalized and verified	● Succeeded
`run_7f35`	billing-prod	Table change · invoices	`billing-changes r4`	Finalized and verified	● Succeeded

← Overview

New backup policy

A policy defines what is captured, what starts it, where copies go and how long they are kept. Every run records an immutable snapshot of the policy revision it used.

01
Source

Policy name: nightly-orders

Source connection: orders-prod · PostgreSQL 17

PROD · ● Connection reachable · ● Read privileges sufficient · ● Tooling version supported

02
What to capture

Selection

- ○ **Whole database** — Includes objects added later
- ○ **Schemas** — Fixed list or a name rule
- ◉ **Selected objects** — Fixed selection, resolved per run

`public` schema

- ☑ `orders` table
- ☑ `order_items` table
- ☑ `customers` added: referenced by orders
- ☐ `products` table
- ☐ `daily_revenue` materialized view
- ☑ `set_updated_at()` function

Added automatically: `customers`, sequence `orders_id_seq`, indexes and constraints on selected tables. Needed outside this backup: extension `pgcrypto` must already exist on the restore target.

Content: Schema and data

Consistency: Single snapshot, no table locks

03
What starts a backup

☑ **Schedule**

Repeat: Every day at 03:00 | **Time zone**: Asia/Jakarta (UTC+07: | **If the runtime was down**: Run once, then resum

☑ **Table change** orders, order_items

Starts a backup when a change is observed. It is not incremental and does not replace log-based recovery. Needs change-history access on the source; if that is lost, protection shows as degraded.

☐ **Database change**

☐ **Webhook** Authenticated, replay-protected

Effective policy

Captures
5 objects from `orders-prod`: 3 tables, 1 function, 1 sequence. Schema and data, one snapshot.

Starts when
Daily at 03:00 Jakarta time, or after a change to `orders` / `order_items` goes quiet for 30 s.

If another event arrives
It merges into the waiting run. At most one run per 15 minutes.

Output
SQL files, one per table, plus manifest.

Complete when
Local and S3 copies are finalized and checked.

● All required fields valid

No restore test yet: run one after the first backup

Save and run first backup

Save without running

- [ ] **MQTT topic**

- [ ] **Redis**

When events arrive in bursts

Wait for quiet (seconds)	Never defer longer than (minutes)	Minimum time between runs (minutes)
30	10	15

Events inside the wait window merge into one run. Merged and skipped events are listed in run history with the reason.

04

Output format

- ( ) **Single SQL file**
  One executable PostgreSQL SQL payload. Restore order and prerequisites are documented in the manifest.
- (•) **Multiple SQL files, by table**
  Shared definitions stored once, with restore order and dependency metadata.
- ( ) **Encrypted SimaX binary**
  Versioned and authenticated. Restore needs the passphrase or key, held separately. If it is lost, the backup cannot be recovered.

05

Where copies go

- **Local filesystem** `/var/backups/simax` — Required
- **AWS S3** `orders-backups/nightly/` — Required
- **SFTP** `archive.example.internal` — Optional

Add a destination: Local | FTP | SFTP | AWS S3 | Google Cloud Storage

A backup is complete only when every required copy is finalized and checked. An optional copy that fails is shown, but does not fail the backup.

06

Retention

Keep for (days)	Always keep at least (usable copies)
30	3

Pinned backups and backups that others depend on are never cleaned up. Deletions are previewed first and written to the audit log.

← Overview

Restore

Nothing on the target changes until you review the preview and confirm. Checks run against the artifact and the target before any restore starts.

01
Artifact

orders-prod · 19 Oct 2026, 03:00
Recovery point from snapshot time, not upload time · `run_7f37` · policy `nightly-orders r11`
PostgreSQL 17 · multiple SQL files, by table

Choose another

● Complete, 2 of 2 copies ● Integrity verified 03:06 This artifact never restore-tested

Retrieve from

Local filesystem · /var/backups/simax

02
What to restore

- ○ **Everything in the artifact**
- ◉ **Selected objects**
  orders, order_items

Dependencies added to the selection: `customers` (referenced by `orders.customer_id`) and sequence `orders_id_seq`. Remove them and the restore is blocked, unless you accept a partial restore.

03
Restore target

- ◉ **orders-restore-test** `DISPOSABLE`
  PostgreSQL 17.2 · different server from the source · recovery drill
- ○ **orders-prod** `PROD`
  Original server. Existing data will be affected: see the preview below.

Checks on this artifact and target

- ● **Artifact complete and readable**
  All 5 files present; checksums match the manifest.
- ● **Engine and version compatible**
  Source PostgreSQL 17, target 17.2.
- ● **Encoding and collation match**
- ▲ **Extension `pgcrypto` is missing on the target**
  Needed by `customers`. Install it on the target, then run the checks again.
- ■ **Role `orders_app` does not exist on the target**
  Owner of 3 selected objects. Map them to another role below, or create the role first.

Map owner `orders_app` to

Choose a role on the target

Target schema

public

Ready to restore?

■ Blocked: 1 problem, 1 warning

1. Map the owner role `orders_app`.
2. Install `pgcrypto` on the target, or accept a partial restore.
3. Resolve the `customers` conflict, or choose new names.

Stages
Retrieve · Verify · Prepare · Replay · Validate

After it finishes
Object list and data checks are compared with the manifest, and the result is saved as restore-test evidence for this artifact.

Run restore on orders-restore-test

Run checks again

04

If the target already has these objects

- (•) **Stop on conflict** Default
  Restore only into names that are free. Nothing existing is changed.
- ( ) **Restore under new names**
  Adds a suffix such as `orders_restored`.
- ( ) **Replace existing objects**
  Destructive. Requires typing the target and scope below.

OBJECT	ON TARGET NOW	RESULT
`public.orders`	Not present	Created
`public.order_items`	Not present	Created
`public.customers`	Exists	▲ Conflict: stops

Restored functions and triggers can run code and have effects outside the database. Use a disposable target for first use. Restore is not guaranteed to roll back as a whole if it fails part-way.

← Overview

run_7f3a

Failed at S3 copy

ledger-prod PROD · Oracle Database 19c · started by the 03:00 schedule · policy `nightly-ledger r6` · correlation `corr_c19e`

Stages

Done	Done	Done	Failed	Not run
Resolve scope	**Export**	**Local copy**	**AWS S3 copy**	**Integrity check**
03:00:04	Snapshot taken 03:00:09	Finalized	Required · 3 attempts	Waits for all required copies

What happened

Affected destination
AWS S3 · `ledger-backups/nightly/` · required

Cause, as reported by S3
Access denied when writing to the prefix. The stored credentials do not have permission to upload there.

Credentials and row contents are never shown in run details.

Safe next step

1. Fix upload permission for the S3 credentials, or edit the destination.
2. Test the destination. This writes and deletes a small marker file.
3. Retry only the S3 copy. The exported files are kept locally and are not regenerated, and nothing already stored is overwritten.

Test destination | Retry S3 copy | Run a full new backup

Recovery point impact

Last usable recovery point
18 Oct 03:00 · 32 h old

This run's files stay on the local disk but are not offered for restore until every required copy is finalized and checked.

See ledger-prod artifacts

Timeline

TIME	EVENT	DETAIL
03:00:00	Schedule occurrence eligible	Daily at 03:00 · Asia/Jakarta
03:00:01	Queued	No other run active on this source
03:00:04	Started	Policy revision r6 recorded for this run
03:00:09	Snapshot taken	This is the recovery point time
03:07:41	Local copy finalized	All files listed in the manifest
03:07:44	S3 upload, attempt 1 of 3	Access denied
03:12:44	S3 upload, attempt 3 of 3	Access denied. Retries exhausted; alert sent

Artifacts

Every backup, with where its copies are and what has actually been checked. A recovery point is only as trustworthy as its latest restore test.

Import artifacts

Source: All sources | Evidence: Any | Format: Any

RECOVERY POINT	SOURCE	FORMAT	COPIES	INTEGRITY	RESTORE TEST	
19 Oct 03:00 `run_7f37`	orders-prod	Multiple SQL, by table	Local, S3 2 of 2 required	Verified	Never tested	Restore
16 Oct 03:00 `run_7e90` Pinned	orders-prod	Multiple SQL, by table	Local, S3 2 of 2 required	Verified	Restored and checked 16 Oct · `drill_0412`	Restore
19 Oct 03:00 `run_7f3a`	ledger-prod	Encrypted SimaX binary	Local only 1 of 2 required	Not checked	Never tested	Incomplete See run
18 Oct 03:00 `run_7e9b`	ledger-prod	Encrypted SimaX binary	Local, S3 2 of 2 required	Verified	Test is 30 days old	Restore

Recovering on a new installation

This catalog is a convenience, not a dependency. On a fresh install, point SimaX at a folder, SFTP path or bucket and it rebuilds the list from the manifests stored beside each backup.

Encrypted backups also need their passphrase or key, which is held outside SimaX. If it is lost, those backups cannot be opened.

Sources

Database servers SimaX reads from or restores to. Each connection is checked before any policy can use it.

Add connection

CONNECTION	ENGINE	ENVIRONMENT	USED AS	PREFLIGHT
orders-prod	PostgreSQL 17	PROD	Source	Ready
billing-prod	MySQL 8.4	PROD	Source	▲ 1 issue
ledger-prod	Oracle Database 19c	PROD	Source	Ready
crm-staging	SQL Server 2022	STAGING	Source	Ready
orders-restore-test	PostgreSQL 17.2	DISPOSABLE	Restore target	Ready

billing-prod PROD

Run preflight again

Host: db-billing.example.internal
Port: 3306
Backup user: simax_backup
Environment label: Production

The password is stored by SimaX and never shown again. The label controls the warnings you see before a restore.

- **Server reachable, identity verified over TLS**
- **Version supported**
  MySQL 8.4. Export tools found on the runtime host.
- **Can read tables, views, routines and events**
  Stored routines are included in backups.
- ▲ **Cannot read change history**
  Needed for table-change triggers. Policy `billing-changes r4` shows as degraded until the account is granted this privilege. Schedules and manual runs are not affected.
- **No changes needed on the source**
  SimaX only reads. It installs nothing on this server.

To fix the warning, ask the database administrator to grant change-history access to `simax_backup`. This changes privileges on the source, so SimaX does not do it for you.

Settings

Encryption keys

Encryption keys | Alerts and roles

Keys protect backups in the encrypted SimaX format. SimaX needs them to restore, but a copy kept only inside SimaX is not a recovery plan.

If every copy of a key is lost, the backups it protects cannot be opened. Nobody can recover them, including SimaX. Keep a recovery copy somewhere separate from the backup storage, and confirm it works.

`ledger-key-2026` ● Recovery copy confirmed

Confirm recovery copy again | Rotate key

Protects	Type	Created	Recovery copy last confirmed
Policy `nightly-ledger` on ledger-prod	Passphrase	2 Jan 2026	12 Oct 2026, by decrypting `run_7e0c`

`reports-key` ▲ Recovery copy not confirmed

Confirm recovery copy | Rotate key

Protects	Type	Created	Recovery copy last confirmed
Policy `reports-weekly` on reports-dev	Key file	4 Mar 2026	Never

Confirm a recovery copy

Check the copy you stored outside SimaX by using it to open a small test backup. The passphrase is used once and not saved.

Key

reports-key

Test backup

Latest reports-dev backup, 15 Oct

Key file or passphrase from your recovery copy

Open test backup

Success is recorded with the date and the backup used. It does not prove that every older backup can be opened.

When you rotate a key

1. New backups use the new key.
2. Older backups still need the old key. Keep it, and its recovery copy, until those backups have expired.
3. SimaX asks you to confirm a recovery copy of the new key.

The file format and a standalone recovery tool are published openly, so backups can be opened without a SimaX installation.

See which backups use each key

Audit log

Who changed protection, ran a restore, handled a key or deleted a backup, and what happened. Entries never contain passwords, keys or database rows.

Export log

Who Anyone

Kind of action All actions

When Last 14 days

TIME	WHO	ACTION	SCOPE	RESULT	CORRELATION
19 Oct 10:41	Retention rule	Deleted 2 expired artifacts Previewed on 18 Oct. Pinned and minimum-copy backups kept.	reports-dev · Local	Done	`corr_b207`
16 Oct 14:20	Operator A	Restored selected objects Recovery drill `drill_0412`	orders-prod backup to orders-restore-test	Done and checked	`corr_a91d`
16 Oct 09:12	Admin B	Changed policy Retention 14 to 30 days · `r10` to `r11`	nightly-orders	Saved	`corr_a6f0`
12 Oct 09:15	Admin B	Confirmed key recovery copy	ledger-key-2026	Done	`corr_9c41`
12 Oct 09:02	Admin B	Edited connection Environment label changed	billing-prod	Saved	`corr_9c3e`
11 Oct 17:30	Webhook	Rejected a request Event ID already used	crm-on-deploy	Denied	`corr_9a02`
9 Oct 15:48	Operator A	Restore to production blocked Role not allowed to restore to a PROD target	orders-prod	Denied	`corr_8e77`

Audit entries are kept for the period set in Settings. Run-level detail, including stages and retries, is in Runs.

Settings

Alerts and roles

Encryption keys | Alerts and roles

Alerts

Tell people when protection gets worse, not only when a run fails.

WHEN	EMAIL	OUTGOING WEBHOOK
A required copy fails	☑	☑
A schedule is overdue	☑	☐
Change observation has a gap	☑	☐
Latest recovery point is older than its limit	☑	☑
Last restore test is older than its limit	☐	☐

Recovery point limit (hours): 26

Restore test limit (days): 30

Email recipients: ops-team@example.internal

Limits apply to every source. A per-source limit can be set on its policy. Alerts never include credentials, keys or row contents.

Roles

Permissions are checked when an action runs, including requests that come in through webhooks and message brokers.

CAN	ADMINISTRATOR	OPERATOR	READ-ONLY
See status, runs and artifacts	Yes	Yes	Yes
Run a backup now	Yes	Yes	No
Restore to a non-production target	Yes	Yes	No
Restore to a production target, or replace objects	Yes	No	No
Edit policies, connections and destinations	Yes	No	No
Manage keys and recovery copies	Yes	No	No
Delete backups and change retention	Yes	No	No
Read the audit log	Yes	Yes	Yes

Keep audit entries for: 1 year

Keep run logs for: 90 days

The roles above are a starting point. Sign-in method and who can hold each role are still to be decided.